

LEARNING RESOURCE GUIDE
ISC2 CCSP: Certified Cloud Security Professional (Aug 2026) | Professional Participant
Workbook

Data Discovery: Structured, Unstructured, and Semi-structured Data

1. Learning Objectives

- By the end of this lecture, you will be able to: Define data discovery and explain why it is a prerequisite for data classification, governance, and protection in cloud environments.
- By the end of this lecture, you will be able to: Distinguish between structured, unstructured, and semi-structured data, and describe the discovery approach and tooling appropriate for each category.
- By the end of this lecture, you will be able to: Explain why cloud environments make data discovery harder than traditional on-premises environments, citing specific characteristics such as dynamic storage, service proliferation, and ephemeral data.
- By the end of this lecture, you will be able to: Describe how automated discovery tools such as Amazon Macie and Microsoft Purview function and what outputs they produce.
- By the end of this lecture, you will be able to: Articulate why data discovery must be a continuous process rather than a one-time project, and explain the role of data ownership in contextualising discovery findings.

2. Introduction

A global investment bank began a cloud migration programme with a stated goal of moving 80% of its workloads to cloud infrastructure within three years. Early in the programme, the compliance team asked a straightforward question: which of the data sets being migrated contain personal data subject to GDPR? The project team's answer was a spreadsheet of known data stores — systems they were aware of and had documented. What they had not accounted for was everything else: the test environment built four years earlier that had been seeded with real customer records, the analytics pipeline that had written intermediate files to cloud storage during a troubleshooting session and never cleaned them up, the archive of email attachments from a decommissioned file server that someone had bulk-uploaded to a cloud storage bucket 'for safekeeping.' The spreadsheet was accurate as far as it went. It just did not go far enough.

Data discovery is the systematic process of finding, identifying, and cataloguing data across an environment. In cloud settings, the challenge is that data accumulates in more places, through more mechanisms, and with less human involvement than in traditional data centres. Storage is cheap and provisioning is fast — which means data spreads quickly, in formats that vary widely, across services that were never intended to be permanent data repositories. Before any classification scheme, any data loss prevention control, or any retention policy can be applied, an organisation needs to know what it actually has and where it actually is.

The three data categories — structured, unstructured, and semi-structured — are not just a taxonomy for exam purposes. They describe meaningfully different discovery challenges that require different tools and techniques. Understanding the distinction is the foundation for designing a discovery programme that actually finds what it is looking for.

3. Core Concepts Explained

3.1 Structured Data

Structured data lives in a well-defined, queryable format — most commonly relational databases organised into tables, rows, and columns with an enforced schema. Customer records in a CRM, transaction records in a billing database, employee records in an HR system, and patient records in a clinical database are all examples of structured data.

Business relevance: Structured data typically contains the most sensitive, high-value information in an organisation — personal identifiers, financial records, health data. It is also the most likely to be subject to regulatory requirements that specify exactly what data must be protected and how.

Discovery approach: The schema itself provides useful signals. A column named 'credit_card_number' or 'date_of_birth' is a direct indicator of sensitivity, even before examining the actual data values. Discovery tools connect to databases, query schemas and metadata, and then apply pattern matching (regular expressions, data type signatures) against sample or full data values to identify sensitive categories: payment card numbers (matching PCI DSS patterns), social security numbers, passport numbers, health record identifiers. Structured data discovery is generally the fastest and most accurate of the three categories.

Real-world example: AWS Macie, when pointed at an RDS database or data stored in structured formats in S3, can identify columns matching sensitive data patterns and generate findings that map to specific regulatory categories such as PII, financial data, or health information.

Risks: Shadow databases — database instances created outside the official inventory for development, testing, or analytics purposes — are a persistent problem. Test environments are frequently seeded with production data and then forgotten. Discovery must include these informal data stores, not just the systems documented in the CMDB.

3.2 Unstructured Data

Unstructured data is everything that does not conform to a predefined schema: emails, word processing documents, PDFs, spreadsheets saved outside a database, presentations, images, audio and video recordings, scanned documents. It represents the majority of data in most organisations by volume, and it is the hardest to discover and classify.

Business relevance: Sensitive information appears in unstructured data frequently and unpredictably — a contract that lists personal data, a scanned ID document stored as a JPEG, an email thread containing account numbers, a presentation with embedded financial projections. The absence of a schema means there is no shortcut to identifying what is inside a file without reading it.

Discovery approach: Content extraction — reading the text inside files — is the only reliable method. Discovery tools for unstructured data use a combination of: (1) format-specific parsers to extract text from PDFs, Office documents, images via OCR; (2) regular expression matching to identify structured patterns within unstructured text, such as credit card numbers or social security numbers; (3) natural language processing to identify sensitive entity types — personal names combined with other identifiers, medical terminology, legal clause language. This is computationally intensive and requires careful tuning to balance recall (finding everything) against precision (avoiding false positives).

Real-world example: A large law firm migrating its document archive to a cloud storage service discovered, through a content inspection scan, that thousands of client files contained embedded personal data — names, addresses, financial account numbers — that had not been identified in the data inventory because the files were stored as opaque PDFs and had never been individually reviewed.

Risks: Unstructured data stores — file shares, email archives, object storage buckets — grow continuously and are rarely pruned. Sensitive data accumulates without any classification or governance. Without active discovery, it is invisible to security controls.

3.3 Semi-structured Data

Semi-structured data has internal organisation — metadata tags, key-value pairs, hierarchical fields — but does not conform to a rigid, schema-enforced format. Common examples include JSON documents, XML files, log files, message queue payloads, API responses, and CSV files where the structure is conventional rather than enforced.

Business relevance: Semi-structured data is everywhere in cloud-native architectures. Application logs, event streams, API interchange data, configuration files, and monitoring telemetry are all semi-structured. These data streams may contain sensitive information — IP addresses, user identifiers, authentication tokens, transaction references — embedded within otherwise routine operational data.

Discovery approach: Semi-structured discovery requires format-aware parsers before content inspection can begin. A JSON document must be parsed to identify its key-value pairs; a log file must be parsed to extract its fields; an XML document must be traversed to reach its data elements. Once parsed, the same pattern matching and entity recognition techniques used for unstructured data apply. The challenge is that semi-structured data often exists in high-velocity pipelines — streaming platforms, message queues — where scanning must be done in near-real-time or on captured samples rather than static files.

Real-world example: A financial services firm's application logging pipeline was found, through a discovery review, to be writing authentication tokens to its operational logs in plaintext. The logs were being shipped to a cloud logging service and retained for 18 months. The tokens were technically valid for only 60 minutes after issuance, but their presence in long-term logs represented a sensitive data exposure that had not been identified without an active discovery scan.

4. Data Category Comparison

Attribute	Structured	Unstructured	Semi-structured
Definition	Fixed schema; tables, rows, columns	No predefined schema; free-form content	Partial organisation; key-value, hierarchical
Common formats	SQL databases, CSV with strict schema	PDFs, emails, images, Word docs, video	JSON, XML, log files, message queues, CSV
Discovery method	Schema inspection + pattern matching on values	Content extraction (OCR, text parsing) + NLP	Format-aware parsing + pattern matching
Discovery speed	Fast — schema provides shortcuts	Slow — every file must be inspected	Medium — parsing overhead before inspection
Common cloud locations	RDS, Aurora, Redshift, BigQuery	S3 buckets, Azure Blob Storage, SharePoint	CloudWatch Logs, Kafka, EventBridge, API GW logs
Primary risk	Shadow databases with production data	Undiscovered sensitive files in object storage	Sensitive tokens/PII in operational logs
Cloud discovery tool	AWS Macie (S3 structured data), DLP	AWS Macie, Microsoft Purview, Varonis	Custom parsers + DLP; stream inspection

5. Case Study

Case Study: The Test Environment Data Contamination Pattern

Financial services and healthcare organisations migrating to cloud have repeatedly encountered a specific discovery finding: test environments built years earlier were seeded with real customer records to make testing realistic, and then left running long after the original project concluded. In several documented cases, these test environments held hundreds of thousands of records — including names, account numbers, and in healthcare contexts, diagnosis codes and treatment histories — with no access controls beyond the basic cloud account permissions.

The data had never appeared in any data inventory because it was stored in infrastructure classified as 'development' rather than 'production,' and the assumption was that development environments did not contain sensitive data. Discovery scans revealed otherwise. In one healthcare case, a formal HIPAA risk assessment had to be conducted to determine whether the exposure constituted a reportable breach, even though the data had not been actively exfiltrated.

The remediation required deleting the data, notifying privacy counsel, and redesigning test data generation practices to use synthetic data rather than production records.

The lesson: data inventory documentation describes what teams intended to build. Data discovery reveals what actually exists. The gap between those two things is almost always larger than expected, and the most significant sensitive data exposures are often in the places that nobody thought to look.

6. Memory Anchor

Think of it this way: Think of structured data as a library with a card catalogue — each book is in a specific location and labelled. Unstructured data is a warehouse full of unmarked boxes — the only way to know what is in each box is to open it and look inside. Semi-structured data is like boxes that are partially labelled — you know roughly what category of contents to expect, but you still need to open the box to find the specific sensitive items. Data discovery is the process of opening all the boxes.

7. Discovery as a Continuous Process

A point-in-time data discovery exercise produces a snapshot. In a cloud environment, that snapshot begins aging immediately. Storage resources are provisioned without central approval. Applications write data to new locations during scaling events. Migrations move data between services. Analytics pipelines produce intermediate files that persist beyond their intended lifespan. An inventory completed six months ago in a dynamic cloud environment may be materially inaccurate today.

Effective discovery programmes use scheduled automated scanning — running on a defined cadence, often daily or weekly for high-priority storage classes — combined with event-driven alerts when new storage resources are created or when sensitive data appears in unexpected locations. The goal is not to run a discovery project; it is to maintain a continuously current understanding of what sensitive data exists and where.

Tooling like Amazon Macie supports this through automated, ongoing scanning of S3 buckets, with findings pushed to security event management systems. Microsoft Purview provides similar capabilities across Azure storage services. These tools shift discovery from a periodic audit function to an operational security control.

A critical nuance: discovery tools identify that sensitive data exists in a location. They cannot determine whether that data is there intentionally. An automated finding that a bucket contains PII could mean it is a legitimate data store with appropriate controls, an orphaned file from a project that was never cleaned up, or an unintended exposure from a misconfigured application. Resolving that ambiguity requires data ownership — a named person responsible for each data store who can explain why the data is there, whether it is supposed to be, and what governance applies to it. Discovery without ownership produces findings that cannot be acted on.

8. Exam Focus

The ISC2 CCSP examination tests the ability to apply data discovery concepts to cloud scenarios, not to recall tool names or implementation specifics.

- Know the three data categories and be able to identify which category a given data type falls into from a scenario description.
- Understand that discovery is a prerequisite for classification — you cannot classify data you have not found.
- Know that unstructured data requires content inspection because there is no schema to provide shortcuts.
- Understand the difference between discovering that data exists and understanding its context — the latter requires data ownership processes.
- Know that cloud characteristics (dynamic provisioning, automatic replication, ephemeral data) make discovery harder and make continuous scanning necessary.
- Common distractor: exam questions may imply that a completed data inventory is equivalent to data discovery. It is not — an inventory reflects what was documented, not what exists.

9. Key Terms Glossary

Data discovery: The systematic process of identifying, locating, and cataloguing data across an environment to determine what exists and where.

Structured data: Data organised in a fixed, schema-enforced format, typically stored in relational databases with defined rows, columns, and data types.

Unstructured data: Data with no predefined schema or fixed format, such as emails, documents, images, and multimedia files.

Semi-structured data: Data with partial organisation through metadata tags, key-value pairs, or hierarchical fields, but without a rigid enforced schema; examples include JSON, XML, and log files.

Content inspection: The process of reading the contents of a file or data object to identify sensitive information, as opposed to relying solely on metadata or schema information.

Shadow database: A database instance created outside official governance processes, often for development or testing purposes, that may contain sensitive production data.

Amazon Macie: An AWS managed service that uses machine learning to automatically discover, classify, and protect sensitive data stored in Amazon S3.

Microsoft Purview: A Microsoft data governance and compliance service that provides data discovery, classification, and protection capabilities across Azure and Microsoft 365 environments.

Data ownership: The assignment of a named individual as responsible for a specific data store, including explaining its contents, governance, and access controls.

OCR (Optical Character Recognition): Technology that converts images of text into machine-readable text, enabling content inspection of scanned documents and image files.

Synthetic test data: Artificially generated data that mimics the statistical properties of production data without containing real personal information, used to eliminate the risk of production data contamination in test environments.

10. Quick Revision Checklist

- ☐ I can define data discovery and explain why it must precede classification and governance.
- ☐ I can describe the characteristics of structured, unstructured, and semi-structured data and give examples of each.
- ☐ I understand that unstructured data requires content inspection because there is no schema to guide discovery.
- ☐ I can explain why cloud environments make data discovery more challenging than traditional on-premises environments.
- ☐ I know that data discovery must be continuous, not a one-time project, due to the dynamic nature of cloud data.
- ☐ I understand the difference between a data inventory and data discovery — and why inventories are insufficient on their own.
- ☐ I can name at least two cloud-native discovery tools and describe what they are designed to scan.
- ☐ I know that test environments seeded with production data represent a common sensitive data exposure pattern.
- ☐ I understand that discovering data in a location does not explain why it is there — data ownership provides the context.
- ☐ I can explain the role of data discovery as the foundation of a data security programme.

11. Practice Questions

Question 1 (Scenario-Based)

A healthcare organisation completes a cloud migration risk assessment and concludes that no PHI (Protected Health Information) exists in its cloud development environment. Three months later, an automated discovery scan using Amazon Macie identifies 47,000 files containing names, dates of birth, and diagnosis codes in an S3 bucket tagged as 'dev-archive.' The data matches records from the organisation's EMR system from three years prior. Which of the following BEST explains why the risk assessment missed this data?

A. The risk assessment relied on documented data inventories rather than active scanning of actual data stores.

→ *Correct. Risk assessments that rely on documented inventories only identify data that someone knew about and recorded. Active scanning through discovery tools finds data regardless of whether it was documented — including data that was moved, copied, or forgotten without updating the inventory.*

B. Amazon Macie was not configured to scan development environment buckets.

→ *Incorrect as a causal explanation. Even if Macie was not configured for dev buckets, this describes a tooling gap, not the root cause of why the assessment failed. The assessment methodology — relying on documentation rather than scanning — is the fundamental issue.*

C. The S3 bucket lacked appropriate data classification tags.

→ *Incorrect. The absence of classification tags is a symptom of the governance gap, not the reason the risk assessment missed the data. Even well-tagged resources require active discovery to verify that the tag accurately reflects the contents.*

D. PHI is not typically stored in cloud environments and therefore was not included in the assessment scope.

→ *Incorrect. PHI is frequently stored in cloud environments, including in development and test environments where it should not be. Excluding cloud from PHI assessment scope is itself a risk.*

Question 2 (Scenario-Based)

A financial services firm's security team reviews the output of an automated data discovery scan. The scan identified 12 S3 buckets containing files with patterns matching credit card numbers. The team escalates the finding to the data governance team, who determine that 9 of the 12 buckets are legitimate payment processing archives with appropriate encryption and access controls, 2 are orphaned from a terminated project and contain live card data, and 1 is a test environment bucket seeded with real card data. Which finding represents the HIGHEST immediate security risk?

A. The 9 legitimate payment processing archives.

→ *Incorrect. These buckets have appropriate controls in place. While they warrant ongoing governance, they do not represent an immediate security risk.*

B. The 2 orphaned project buckets containing live card data.

→ *Correct. Orphaned buckets from terminated projects typically have degraded access controls, may not be monitored, and may have had their access policies changed over time. Live card data in unmonitored, ungoverned storage represents a high-probability exposure risk. These require immediate remediation.*

C. The 1 test environment bucket seeded with real card data.

→ *Partially correct — this is a serious finding — but the 2 orphaned buckets with live data and no active ownership or monitoring represent a higher immediate risk. At least the test environment is ostensibly under active development oversight.*

D. All 12 buckets equally, since all contain card data patterns.

→ Incorrect. The risk profile of each bucket differs based on the adequacy of controls, the governance status of the bucket, and whether ownership is clear. Risk triage, not uniform treatment, is the correct approach.

Question 3 (Scenario-Based)

A cloud security architect is designing a data discovery programme for a retail organisation that stores customer transaction data in an Aurora relational database, customer support email archives in an S3 bucket, and real-time clickstream data in a Kinesis Data Stream. The architect needs to define discovery approaches for each data store. Which combination of discovery methods is MOST appropriate?

A. Schema inspection with pattern matching for Aurora; document tagging policies for the S3 email archive; no scanning needed for Kinesis as streaming data is transient.

→ Incorrect. Document tagging policies do not discover what data is inside files — they manage metadata. Kinesis data is transient but may contain sensitive information, and sampling-based inspection is appropriate.

B. Schema inspection with pattern matching for Aurora; content extraction and pattern matching for the S3 email archive; near-real-time or sample-based content inspection for the Kinesis stream.

→ Correct. Each method is matched to the data type. Relational databases use schema shortcuts plus value scanning. Email archives require reading file contents. Streaming data requires inspection of the data payload, either in near-real-time or through representative sampling.

C. Pattern matching on all three data stores using the same tool configuration.

→ Incorrect. A single tool configuration cannot appropriately handle all three formats. Email archives require content extraction before pattern matching; Kinesis data requires stream-aware inspection. Applying a generic pattern matching configuration to all three will produce poor recall for the unstructured and semi-structured stores.

D. Schema inspection for Aurora; encryption key scanning for S3; rate-limiting inspection for Kinesis.

→ Incorrect. Encryption key scanning is not a data discovery technique. S3 email archives require content inspection. This option describes a combination of irrelevant techniques.

Question 4

Why is unstructured data the most challenging data category for discovery?

A. Unstructured data is always encrypted, which prevents scanning tools from reading its contents.

→ Incorrect. Encryption status is independent of data structure category. Many unstructured files are not encrypted, and encryption can be applied to any data type. This is not why unstructured data is harder to discover.

B. Unstructured data has no predefined schema, so every file must be individually inspected to determine whether it contains sensitive information.

→ *Correct. Without a schema, there are no field names, column types, or structural signals to guide discovery. The only way to determine whether an email, document, or image contains sensitive data is to read its contents, which is computationally intensive and must be done on every file individually.*

C. Unstructured data cannot be stored in cloud environments and therefore falls outside cloud discovery tooling.

→ *Incorrect. Unstructured data is extensively stored in cloud environments — in object storage services, file storage, email platforms, and content management systems. It is very much within scope for cloud discovery.*

D. Unstructured data changes too frequently for any discovery tool to maintain an accurate inventory.

→ *Incorrect. While unstructured data stores do change frequently, discovery tools can scan on a scheduled basis and produce current findings. Frequency of change is a reason to run discovery continuously, not a reason why it is impossible.*

Question 5

Which of the following is the BEST description of semi-structured data?

A. Data that has been partially classified by a governance tool but not yet fully tagged.

→ *Incorrect. This describes a state in a data governance workflow, not a data format category. Semi-structured is a description of data format, not classification status.*

B. Data with internal organisation through metadata, key-value pairs, or hierarchical fields, but without a rigid enforced schema.

→ *Correct. Semi-structured data — JSON documents, XML files, log entries, message queue payloads — has enough internal structure to be parsed but lacks the strict schema enforcement of a relational database.*

C. Data stored in both relational databases and file systems simultaneously.

→ *Incorrect. This describes multi-location storage, not a data category. The location of data does not determine whether it is structured, unstructured, or semi-structured.*

D. Data that has been anonymised and can therefore be treated as less sensitive.

→ *Incorrect. This describes a data processing state (anonymisation), not a structural category. Semi-structured data can be highly sensitive.*

Question 6

A data governance team has completed a formal data discovery exercise and documented every sensitive data store in a comprehensive inventory. What is the PRIMARY limitation of treating this as a completed project rather than an ongoing programme?

A. The discovery tools used may have missed some data due to configuration errors.

→ *Incorrect. While tool configuration errors are a valid operational concern, they are not the primary limitation of treating discovery as a one-time project. Even a perfectly executed point-in-time discovery exercise becomes outdated immediately.*

B. Cloud environments are dynamic — new data is created, moved, and replicated continuously, so an inventory completed once becomes inaccurate quickly.

→ *Correct. Cloud environments generate data continuously through new storage provisioning, application writes, replication, caching, and automated processes. A point-in-time inventory begins aging from the moment it is completed. Only continuous scanning maintains an accurate picture.*

C. The regulatory requirements for data discovery specify that it must be conducted quarterly, so annual completion is non-compliant.

→ *Incorrect. While some regulatory frameworks do specify discovery frequency, the primary limitation of a one-time approach is technical — the dynamic nature of cloud data — not a specific regulatory calendar requirement.*

D. The data discovered will have changed classification by the time the inventory is reviewed.

→ *Incorrect. Classification changes are a real governance concern but they are secondary to the more fundamental issue: new data stores appear and existing stores change between discovery cycles, making the inventory itself incomplete.*

Question 7

What is the role of data ownership in the context of a data discovery programme?

A. Data owners are responsible for running the discovery scanning tools.

→ *Incorrect. Discovery tooling is typically operated by security or data governance teams, not individual data owners. Data ownership is about accountability for specific data stores, not technical tool operation.*

B. Data owners provide context that explains why sensitive data exists in a specific location, enabling appropriate governance decisions after discovery finds it.

→ *Correct. Discovery tools identify that sensitive data exists in a location. They cannot determine whether it is supposed to be there, who is responsible for it, or what governance applies. Data owners provide the human context that transforms a raw finding into an actionable governance decision.*

C. Data owners must approve every access to sensitive data identified during a discovery scan.

→ *Incorrect. While data owners may be involved in access control decisions, their primary role in the discovery context is to provide accountability and context for findings, not to serve as a bottleneck for every subsequent access.*

D. Data owners are responsible for encrypting the data identified in discovery findings.

→ *Incorrect. Encryption decisions and implementation are typically handled by technical teams following governance policies. The data owner's role is accountability and context, not direct implementation of security controls.*

Question 8

Which cloud-native service is specifically designed to discover and classify sensitive data in Amazon S3 using machine learning?

A. AWS Config

→ Incorrect. AWS Config tracks configuration changes to cloud resources and evaluates them against compliance rules. It does not inspect the contents of data objects to identify sensitive information.

B. Amazon Macie

→ Correct. Amazon Macie is a managed data security service that uses machine learning to automatically discover, classify, and protect sensitive data stored in Amazon S3. It generates findings when it detects patterns matching PII, financial data, health records, and other sensitive categories.

C. AWS GuardDuty

→ Incorrect. GuardDuty is a threat detection service that monitors for malicious activity and unauthorised behaviour using CloudTrail logs, VPC flow logs, and DNS logs. It does not inspect the contents of stored data objects.

D. AWS Security Hub

→ Incorrect. Security Hub aggregates and prioritises security findings from multiple AWS services. It receives findings from services like Macie but does not itself perform data content inspection.

Question 9

An organisation's web application logs are being shipped to a cloud logging service and retained for 24 months. A discovery review finds that the logs contain authentication session tokens in plaintext. What data category do these logs represent, and what is the primary risk?

A. Structured data; the risk is that the log format is not schema-validated.

→ Incorrect. Web application logs are semi-structured data, not structured data. The risk is not about schema validation.

B. Semi-structured data; the risk is that sensitive credentials embedded in operational logs are being retained beyond their useful lifespan in a potentially lower-security logging environment.

→ Correct. Logs are semi-structured (fields with recognisable patterns but no rigid schema). The tokens themselves are valid for only minutes, but their presence in 24-month log archives means they persist indefinitely. Log environments often have broader access than production systems, creating unnecessary credential exposure.

C. Unstructured data; the risk is that the log files are too large to be indexed by discovery tools.

→ Incorrect. Logs are semi-structured, not unstructured. File size is an operational consideration for discovery tooling, not the primary risk.

D. Structured data; the risk is that the tokens could be used to reconstruct database queries.

→ Incorrect. Log files are semi-structured. Authentication tokens do not enable database query reconstruction; they enable session impersonation.

Question 10

A CCSP candidate is asked to explain why a completed data map cannot substitute for active data discovery in a cloud environment. Which explanation is MOST accurate?

A. Data maps use a different file format than discovery tools, making them incompatible.

→ Incorrect. File format compatibility is a technical implementation detail, not a conceptual explanation. This misses the point of the question entirely.

B. A data map documents what was known and documented at a point in time; active discovery finds what actually exists, including data stores and data that were never recorded in the map.

→ Correct. Data maps are based on human documentation — what teams intended to build and what they chose to record. Discovery is based on what actually exists in the environment at the time of the scan. The gap between the two — shadow databases, forgotten test environments, orphaned files — is precisely what makes active discovery necessary.

C. Data maps are produced by compliance teams who lack technical access to cloud storage, so they are always inaccurate.

→ Incorrect. Data maps can be produced by technical teams with full access, and they can be accurate as far as they go. The limitation is not who produced the map but what the map captures — only what was known and documented.

D. Active discovery can only find structured data, making it a complement to the data map rather than a replacement.

→ Incorrect. Active discovery can find structured, unstructured, and semi-structured data. It is not limited to any single data category.